Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Date: Sep 6, 2023 | **Pre-Tutorial Assignment 5** | **Marks: 2**

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1. A linear network **N** (with unique solution) delivers $3.5A$ when the load connected is 5.25Ω and $2.1A$ when the load is 19.5Ω. What is the maximum power $P_{max}(\mathbf{N})$ (in W) that **N** can deliver?

2. Consider an Excess-3 to BCD code converter. Let the Excess 3 input be represented by four bits $PQRS$ and the BCD output be represented by four bits $ABCD$.

   (a) Draw the K-maps for A and B

   (b) Find a Boolean expression for A in the minimal SOP form.

   (c) Find a Boolean expression for B in the minimal POS form.